

Blood Donor Matching Platform

Product direction

Connect people who need blood with eligible nearby donors through a simple request, matching, notification, response, chat, and confirmation flow. Hospitals support trust through request verification, and partner institutions (blood banks and welfare organizations) support fulfillment and can also request blood themselves; neither is the primary user experience.

Prepared for internship-level implementation and evaluation.

Document at a glance

This PRD keeps the original Blood Donor Matching Platform problem and terminology, but reduces the first release to a small set of capabilities that can be built, demonstrated, tested, and defended by an internship team.

Item	Decision
Problem	Blood requests can be difficult to fill quickly because suitable donors are hard to find and coordinate.
Primary users	Requester and Donor
Supporting actor	Hospital verifies requests tied to its own location; Partner Institution (blood bank or welfare organization such as Alkhidmat) can request blood and fulfill other requests
MVP channel	Push notifications only for donors connected to the internet
Core outcome	A request reaches a suitable donor, the donor responds, and the donation outcome is recorded.
Release posture	Practical MVP; manual review is acceptable where automation would add unnecessary complexity.

1. Product context and goals

1.1 Problem statement

When a patient or family needs blood, the requester may not know which donors are available, nearby, and compatible. Informal messages can be slow, fragmented, and difficult to confirm. A hospital may verify a request tied to its own location, and partner institutions (blood banks and welfare organizations) may hold stock to fulfill a request, but many requests begin outside a registered hospital.

1.2 Product goals

- Make it easy for a requester to create a complete blood request, even when their hospital is not registered.
- Help donors control whether they are available and receive relevant requests through internet-based push notifications.
- Provide a lightweight response and coordination path: accept or decline, then simple in-app chat.
- Record whether a donation was completed and close requests clearly.
- Give hospitals a supporting workflow to verify requests tied to their own location.

- Give partner institutions (blood banks and welfare organizations) a workflow to create their own requests and fulfill other requests, alongside individual donors. [NEW]
- Provide basic admin review and measurable evidence that the MVP works.

1.3 Success definition

The MVP is successful when a test request can move from creation to matching, notification, donor response, chat, donation confirmation, and closure with an auditable status history.

Scope guardrail

The first release optimizes for a reliable core workflow, not a complete blood-bank operating system. Manual admin review and simple rules are acceptable for an internship MVP.

2. Users, roles, and trust model

Role	Needs	MVP permissions
Requester	Create and track a request; communicate with a willing donor; confirm outcome.	Register, create/update/cancel own requests, view responses, chat after willingness, confirm fulfillment.
Donor	Control availability; see relevant requests; respond safely.	Register, set blood group/location/availability, view notified requests, accept/decline, chat after acceptance, confirm donation.
Hospital	Verify requests tied to its own location; no fulfillment role.	Register hospital, submit verify/not-verify decision with a reason on requests tied to its location.
Partner institution	Request blood as needed; fulfill other requests from declared stock.	Register institution, create requests, view/respond to open requests, fulfill from declared stock, record staff confirmation.
Admin	Keep the system trustworthy and operational.	Review accounts/requests, manage trust labels, resolve basic flags, view metrics, deactivate access manually.

2.1 Trust labels

Trust labels communicate the source of a request; they do not claim medical or legal certification. Use three simple labels:

Label	Meaning	Display
Self-verified request	Created by a requester without partner approval.	Visible with a clear "Self-verified" label.
Institution-backed request	Reviewed by the requester's own registered hospital before becoming Active; hidden from donors until verified.	Visible with institution name and "Institution-backed" label.
Partner fulfillment	A partner indicates that available stock may fulfill the request.	Visible on request status; final confirmation still comes from the requester or institution.

2.2 Registration and identity assumptions

- Requester and donor register with phone number and OTP; basic profile fields are required.
- Donor blood group is self-declared in MVP and shown as “Donor-provided”; verification can be a future enhancement.
- Hospitals require admin approval before their verification decisions are trusted. Partner institutions require admin approval before their request-creation and fulfillment actions are trusted.
- CNIC or other government-ID capture is optional for the prototype and should not be stored unless a supervisor approves the privacy and security design.

3. MVP scope and user journeys

3.1 In-scope capability map

Register / profile → Create / request → Match / nearby → Notify / push → Respond / chat → Confirm / close

3.2 Requester journey

- Register and complete a basic profile.
- Create a request with blood group, units required, location/hospital, urgency, and required-by date/time.
- If the requester's declared hospital is registered on the platform, the request enters Pending hospital verification and stays hidden from donors until that hospital reviews it; if the hospital is not registered, the request becomes Active immediately with a Self-verified label.
- View matching activity and donor responses; chat becomes available only after a donor accepts or expresses willingness.
- Mark units fulfilled as blood is received, or cancel the request if no longer needed. The request is fully fulfilled only once confirmed units fulfilled equal units required — if only part of the requested quantity is fulfilled, the request stays active for the remaining units until they are fulfilled, the requester cancels, or the request expires. [UPDATED]

3.3 Donor journey

- Register with blood group, location, contact preference, and availability.
- Receive a push notification when a matching request is created — the donor does not need to have the app open or be actively using it at that time. If the donor's device has an active internet connection and notification permission is enabled, the platform can deliver the notification through the device's native notification system while the app is closed or in the background. [UPDATED]
- Open request details and choose Accept / I can donate or Decline.
- After willingness, use in-app chat to coordinate; chat is non-binding and does not guarantee donation.
- After the appointment or handover, confirm Donation completed or indicate that it did not happen.

3.4 Hospital journey

- Register hospital details and wait for admin approval.
- Review incoming requests tied to the hospital's own location and mark them verified or not verified with a short reason.
- Once verified, the request becomes Active and is released to donor matching; the hospital has no fulfillment or stock role. [UPDATED]

3.5 Partner institution journey

- Register institution details and wait for admin approval.
- Create an institution request when stock is low or a patient need is known, using the same request form as an individual requester.
- Browse open requests in its service area and indicate that matching stock is available; the institution competes with individual donors on a first-to-respond basis. [UPDATED]
- Record fulfillment once matched; the requester or institution still confirms the final outcome.

4. Functional requirements

4.1 Accounts, profiles, and availability

ID	Requirement	Priority
FR-01	The system shall support phone-number registration and OTP login for Requesters, Donors, Partners, and Admins.	Must
FR-02	The system shall capture name, phone, role, and consent to the privacy notice.	Must
FR-03	A donor shall provide blood group, approximate location, and availability status: Available, Temporarily unavailable, or Not available.	Must
FR-04	A donor shall be able to turn notifications off without deleting the account.	Must
FR-05	A hospital shall require admin approval before it can verify requests, and a partner institution shall require admin approval before it can create requests or claim stock fulfillment.	Must
FR-06	The system shall show trust labels and the timestamp of the latest relevant verification.	Should

4.2 Request creation and management

ID	Requirement	Priority
FR-07	A requester shall create a request with required blood group, units required, urgency, hospital/location, required-by date/time, and a short note. [UPDATED]	Must
FR-07a	The system shall track Units Required, Units Fulfilled, and Units Remaining for every request. A request is fully fulfilled only when confirmed Units Fulfilled equal Units Required; partial fulfillment does not close the request. [NEW]	Must
FR-08	A requester shall be allowed to submit a request even when the hospital is not registered.	Must
FR-09	A request shall become Active and visible to donors immediately with a Self-verified label if the requester's hospital is not registered; if the hospital is registered, the request shall enter Pending hospital verification and stay hidden from donors until that hospital verifies it, after which it becomes Active with an Institution-backed label.	Must
FR-10	A requester shall view, edit, cancel, and mark their own request fulfilled.	Must
FR-11	The system shall prevent new donor responses after the request is cancelled, fulfilled, expired, or closed.	Must
FR-12	A requester shall see basic status and response counts, not private donor data before willingness.	Must

4.3 Matching and notifications

Matching boundary

The MVP uses a project-approved blood-group compatibility table to determine potential donor matches. This is logistical matching only — the platform does not provide medical advice or make final transfusion-eligibility decisions. Final donor eligibility and transfusion suitability remain the responsibility of the relevant hospital, blood bank, or qualified medical professional. [NEW]

ID	Requirement	Priority
FR-13	The system shall match by compatible blood group, donor availability, and approximate distance from the request location.	Must
FR-14	The system shall rank candidates by compatibility, distance, and availability; exact ranking rules may remain simple and documented.	Must
FR-15	The system shall send push notifications to eligible donors when their device is connected to the internet and notification permission is available. The donor does not need to have the application open or be actively using it when the request is created — if the device has an active internet connection and notifications enabled, the platform may deliver the notification through the device's notification system. [UPDATED]	Must
FR-16	A notification shall open the request details in the app.	Must
FR-17	If no donor responds within a configurable wait window, the system shall notify the next batch or expand the radius within configured limits.	Should
FR-18	The system shall show "Notification delivery not guaranteed" where a device is offline, notifications are disabled, or delivery is unknown.	Must

4.4 Response, chat, and confirmation

ID	Requirement	Priority
FR-19	A donor shall be able to Accept / I can donate or Decline a notified request.	Must
FR-20	The first accepted response may be marked Matched / In progress; additional responses remain visible as backup until the request is fulfilled or closed.	Should
FR-21	In-app chat shall become available only after a donor expresses willingness or accepts.	Must
FR-22	Chat shall support simple text messages, timestamps, and a basic report/flag action.	Must
FR-23	The system shall not treat chat as proof of donation or medical advice.	Must
FR-24	The donor and requester shall be able to record Donation completed (updating Units Fulfilled accordingly), Did not happen, or Cancelled. [UPDATED]	Must
FR-25	A partner institution may record staff confirmation for an institution-backed request.	Should
FR-26	The request shall close after full confirmed fulfillment (Units Fulfilled equal Units Required), requester cancellation, or expiry. [UPDATED]	Must

4.5 Partner support and admin

ID	Requirement	Priority
FR-27	Partner institutions can view relevant blood requests in their registered service area and respond when they can help fulfill a request, competing with individual donors on a first-to-respond basis. [UPDATED]	Should
FR-28	A hospital shall mark a request tied to its location as Verified or Not verified with a reason; a verified request becomes Active and visible to donors, and a not-verified request stays hidden pending resolution.	Should
FR-29	A partner institution shall create its own donor request using the same request form and institution-backed label.	Must
FR-30	Stock fulfillment shall be a simple declaration of availability and fulfillment; no batch-level inventory ledger is required.	Must
FR-31	Admin shall search users and requests, review flags, change trust labels where justified, and deactivate an account manually.	Must
FR-32	Admin shall view basic metrics for requests, matches, responses, confirmations, and notification outcomes.	Should

Notification boundary

MVP push notifications require internet connectivity on the donor device. IVR, SMS, WhatsApp, phone calls, and offline delivery are intentionally not part of this release.

5. Lifecycle, states, and edge cases

5.1 Request state model

Draft → (Pending hospital verification) → Active → Matched / In progress → Fulfilled → Closed

State	Entry condition	Allowed next states
Draft	Requester is completing the form.	Pending hospital verification, Active, Draft discarded [UPDATED]
Pending hospital verification [NEW]	Requester's declared hospital is registered; request is hidden from donors awaiting that hospital's verification.	Active, Closed (rejected)
Active	Required fields pass validation; matching is enabled.	Matched / In progress, Fulfilled from stock, Cancelled, Expired
Matched / In progress	At least one donor has expressed willingness or a partner has claimed stock. Remains in this state while Units Fulfilled is less than Units Required.	Fulfilled, Active after cancellation, Cancelled
Fulfilled	Confirmed Units Fulfilled equal Units Required (requester or partner confirms the need was fully met).	Closed
Closed	Final state after fulfillment, cancellation, or expiry.	None

5.2 Practical edge cases

Situation	MVP behavior
Donor is offline or push notifications are unavailable	Successful push delivery is not assumed; delivery status is shown as unknown and the request remains available for other eligible donors.
Donor has notifications disabled	Do not treat the donor as reachable; exclude from active notification batches until enabled.
No matching donors nearby	Show no-match status, allow radius expansion to a safe configured limit, and let the requester share the request manually.
Two donors accept at nearly the same time	Keep the first confirmed match as primary; mark others as backup or notify that the slot is filled.
Requester cancels after notification	Stop further notifications and close open chat coordination for the request.
Donor accepts then cancels	Reopen the request as Active and resume matching, subject to the request deadline.
Required-by time passes	Mark Expired and stop notifications; retain history for metrics.
Incorrect blood group or location entered	Allow the requester to edit before fulfillment; display a prominent confirmation step before publishing.
Chat contains unsafe or abusive content	Allow either party to flag the conversation; admin reviews manually. No automated suspension in MVP.

6. Non-functional requirements and privacy

The numbers below are reasonable targets for a student prototype, not production service-level commitments.

Area	MVP target / requirement
Usability	A first-time requester should be able to publish a valid request in about 3 minutes in a usability test; donor response should take no more than 2–3 taps after opening the notification.
Performance	For normal test data, request creation and request detail screens should load within 3 seconds on a stable connection. Matching should complete within 10 seconds for a small pilot dataset.
Availability	The demo deployment should be available during agreed testing windows. The app must handle temporary network failure with a clear retry state.
Reliability	No duplicate active request or duplicate donor response should be created by a repeated tap or retry.
Security	Use OTP authentication, role-based access, server-side authorization, encrypted transport, and secure storage for secrets. Never expose donor phone numbers before willingness.
Privacy	Collect only data needed for matching and coordination. Use approximate location for matching/display; do not show an exact home address to other users.
Retention	Keep request and audit history for the project evaluation period; define a deletion path for test accounts and chat data.
Accessibility	Use readable labels, sufficient contrast, clear error messages, and avoid relying only on color for status.
Observability	Log request state changes, match attempts, notification attempts, responses, and errors without logging OTPs or message contents by default.

6.1 Safety and disclaimer basics

- The platform is a coordination tool, not a medical eligibility authority. Donor eligibility and clinical decisions remain with qualified medical staff.
- Show emergency guidance: users should contact the hospital or local emergency services for urgent medical emergencies.
- Do not display sensitive patient diagnoses in public request text.
- Require a consent checkbox before sharing the request and before starting chat.

7. Architecture, data, metrics, and delivery plan

7.1 Simple architecture

Mobile/web UI → API + auth → Matching service → Push provider → Database

Recommended implementation shape: a single backend application with a relational database, a small matching module, and one push notification provider. A modular monolith is sufficient; separate microservices are not required.

Component	Responsibility
Client application	Registration, forms, request views, donor response, chat, status display.
API and authentication	OTP login, role checks, validation, request and profile APIs.
Matching module	Blood-group compatibility, availability, distance calculation, ranking, batch selection.
Notification adapter	Create push notifications and record accepted, attempted, failed, or unknown delivery state.
Database	Users, profiles, institutions, requests, matches, messages, confirmations, flags, and audit events.
Admin console	Basic search, review, trust-label management, manual deactivation, metrics view.

7.2 Minimum data entities

User (id, role, phone, name, consent, status) • DonorProfile (blood group, approximate location, availability) • Institution (name, type: Hospital or Partner institution, location, approval status) • Request (blood group, **units required, units fulfilled, units remaining**, location, urgency, required-by, status, trust label) • Match (request, donor, score, notification status, response) • ChatMessage (conversation, sender, timestamp, text, flag status) • Confirmation (actor, outcome, timestamp) • AuditEvent (actor, action, timestamp). [UPDATED]

7.3 Matching rule for MVP

Filter: donor is Available, blood group is compatible according to the project-approved compatibility table, donor has a usable approximate location, and the request is Active (Units Remaining greater than zero). Rank by compatibility priority, distance, and most recent availability. Notify a small batch first, then continue with the next batch if the wait window ends. Store the score inputs so the result can be explained during evaluation.

This matching is logistical only: it identifies potential donor candidates by blood-group compatibility, availability, and distance. It does not constitute medical advice or a final transfusion-eligibility decision — final donor eligibility and transfusion suitability remain the responsibility of the relevant hospital, blood bank, or qualified medical professional. [NEW]

8. Metrics, delivery plan, and acceptance

8.1 Basic metrics

Metric	Definition
Active requests	Count of requests currently open for matching.
Match rate	Active requests with at least one eligible donor match.
Response rate	Notified donors who accept or decline divided by donors notified.
Time to first response	Time from first notification attempt to first donor response.
Fulfillment rate	Requests marked fulfilled (Units Fulfilled equal Units Required) divided by requests closed, excluding requester cancellations.
Notification outcome	Counts of push attempts, delivered/confirmed where available, failed, and unknown due to offline state.
Partner contribution	Requests verified, institution-created requests, and stock fulfillment records.
Safety flags	Count of flagged users, requests, and chats resolved by admin.

8.2 Internship delivery plan

Iteration	Deliverable
1	Wireframes, data model, roles, OTP stub, and request/profile forms.
2	Donor availability, location capture, basic compatibility and distance matching.
3	Push notification integration, accept/decline, request lifecycle.
4	Chat, donation confirmation, partner support, trust labels.
5	Admin screen, metrics, privacy checks, edge-case tests, deployment and demo script.

8.3 MVP acceptance checklist

- A requester can register and publish a valid request without a registered hospital.
- An available donor with compatible blood group and nearby location is matched and receives a push notification when online.
- The donor can open the request, accept or decline, and chat only after willingness.
- The requester can cancel or fulfill the request, and notifications stop after closure.
- A hospital can be approved and verify a request tied to its own location. A partner institution can be approved, create an institution request, and record simple stock fulfillment.
- An admin can review users/requests, handle a flag, and view the basic metrics.
- Offline donor behavior is visible as unknown/unreachable; no IVR/SMS/offline mechanism is implied.
- Test data can be deleted or anonymized before the project is shared publicly.

9. Scope boundaries and decisions

9.1 Explicitly out of scope for MVP

- IVR, SMS, WhatsApp, automated phone calls, or any offline notification channel.
- Advanced medical eligibility screening, clinical records, prescriptions, or diagnosis data.
- Sophisticated blood-bank inventory management, expiry tracking, batch/lot tracking, or multi-warehouse stock.
- Automated donor scoring, automated suspension, background checks, or a complex dispute-resolution system.
- Payments, rewards, donor marketplace behavior, advertisements, or public social feeds.
- Live map navigation, exact address sharing, and real-time location tracking.
- Complex analytics, predictive demand forecasting, or machine-learning matching.
- Multi-country compliance and production-grade identity verification.

9.2 Future scope / separate decision

These items may be evaluated after the MVP and should be treated as separate product decisions with their own privacy, cost, and operational review:

- Offline escalation through IVR or SMS for donors with cellular service but no internet.
- WhatsApp or call-center coordination where legally and operationally appropriate.
- Verified donor badges, institution accreditation, and stronger identity verification.
- More complete inventory workflows for partner blood banks and hospitals.
- Appointment slots, check-in, reminders, and hospital-side workflows.
- Improved matching rules, multilingual support, accessibility improvements, and richer analytics.

9.3 Key assumptions

- A push notification provider and internet-connected test devices are available for the MVP demonstration.
- The project has a defined pilot geography so distance and partner workflows can be tested realistically.
- A supervisor or designated admin can manually review partners and safety flags.
- The blood-group compatibility rules are reviewed by a qualified project advisor before testing with real users.
- For the internship demo, test data and simulated confirmations are acceptable; real donation operations require institutional approval.

Final product decision

The MVP promises internet-based coordination for registered users. It does not promise that every request will reach a donor, especially when the donor is offline, has disabled notifications, or has no network access.

Appendix A. Requirement traceability

The revised PRD preserves the source concept while narrowing implementation risk:

Source concept retained	MVP interpretation
Requester / Donor / Hospital / Partner institution	Requester and Donor are core users; Hospital and Partner institution are supporting actors with distinct roles (verification vs. request/fulfillment).

Source concept retained	MVP interpretation
Self-verified vs institution-backed	Both labels remain; registration of the hospital is not required to create a request, but a registered hospital now gates visibility until it verifies.
Eligible donor pool	Compatibility + availability + approximate location are the minimum filters.
Match and notify nearest eligible donors	Push notifications are used only for connected donors; delivery uncertainty is explicit.
Accept / I can donate	Donor response is the main conversion event from match to coordination.
Optional in-app chat	Chat is available after willingness and is non-binding.
Donation completed / Request fulfilled / Closed	Confirmation and clear final states remain in the lifecycle; fulfillment now tracked at the unit level.
IVR escalation	Moved to Future Scope / separate decision; excluded from MVP.
Institution stock fulfillment	Kept as a simple support action, not full inventory management.
Admin and metrics	Kept at basic operational level for trust, safety, and evaluation.

Appendix B. Glossary

Term	Meaning
Requester	Person who creates and tracks a blood request.
Donor	Registered person who may respond to a compatible request.
Hospital	Registered hospital that verifies requests tied to its own location; has no fulfillment role. [NEW]
Partner institution	Registered blood bank or welfare organization (e.g. Alkhidmat) that can create its own requests and fulfill other requests; cannot verify requests. [UPDATED]
Self-verified	Request created without partner institution verification.
Institution-backed	Request reviewed by an approved partner institution.
Match	A system-generated association between an active request and an eligible donor.
Willingness	Donor has tapped Accept / I can donate; this is not proof of donation.
Fulfilled	Confirmed Units Fulfilled equal Units Required for the request.
Closed	Final request state after fulfillment, cancellation, or expiry.

End of document